

RATEQUANT: Optimal Mixed-Precision KV Cache Quantization via Rate-Distortion Theory

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Abstract

KV cache quantization reduces the memory footprint of large language model inference, yet existing quantizers assign uniform bit-widths to every attention head, overlooking significant variation in head importance. A natural idea is to allocate more bits to important heads and fewer to the rest. However, we observe that such mixed-precision allocation has a hidden pitfall: each quantizer follows a different distortion curve $D(b) = \alpha\beta^{-b}$, where the decay rate β varies from 3.6 to 5.3 across designs. Applying one quantizer’s distortion model to another inverts the allocation order and makes performance *worse* than uniform quantization, a failure mode we call *distortion model mismatch*. In this work, we propose **RATEQUANT**, a framework that resolves this mismatch by fitting per-quantizer distortion models from a small calibration set, then solving the bit-allocation problem in closed form via reverse waterfilling from rate-distortion theory. Extensive experiments on Qwen3 and Llama3 families across three quantizers demonstrate that calibrated RATEQUANT reduces KIVI’s perplexity at 2.5 bits from 49.3 to 14.9 (70% ↓) and recovers 70–85% of quantization-induced degradation at 4.0 bits, with zero runtime overhead and <2 s one-time calibration.

1 Introduction

Serving large language models (LLMs) at scale requires caching all previously computed key-value (KV) pairs so that each new token can attend to the full context [34, 29]. The memory footprint grows linearly with sequence length, batch size, and model depth, making the KV cache a primary memory bottleneck [30, 15]. KV cache quantization reduces this cost, and recent work has produced effective quantizers [23, 2, 13, 41].

Despite their success, these quantizers apply *uniform* bit-widths to every attention head, implicitly assuming equal contribution from every head. This assumption is increasingly at odds with recent findings. Head importance studies [35, 33] show that heads exhibit highly non-uniform importance. Meanwhile, recent mixed-precision approaches relax uniformity at the layer [17, 18] or channel level [19, 36], but each relies on heuristic rules tied to a specific quantizer. These observations raise a fundamental question: *if heads are not equally important, can we build a principled, quantizer-agnostic framework for mixed-precision KV cache allocation?*

We identify a deeper obstacle that must be resolved first: *distortion model mismatch*. Different quantizers have fundamentally different distortion-rate curves $D(b) = \alpha \cdot \beta^{-b}$, with the decay rate β varying from 3.6 (TURBOQUANT) to 5.3 (QuaRot). As Fig. 1 illustrates, naively applying one quantizer’s distortion model to another makes mixed-precision allocation *worse than uniform*. This

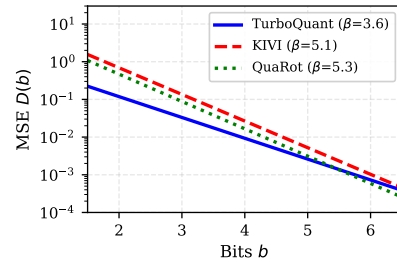


Figure 1: Quantizer β varies 1.5 $\times$; mismatched calibration worsens PPL.

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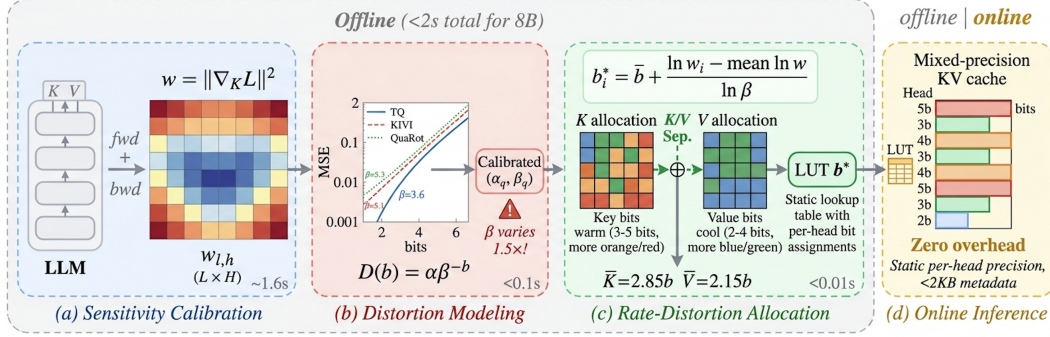


Figure 2: RATEQUANT pipeline. Phases 1–3 are one-time offline costs (<2 s for 8B); Phase 4 adds zero runtime overhead.

occurs because mismatched β inverts the marginal gain ordering across heads, causing the algorithm to allocate bits to the wrong heads. Our investigation is guided by two research questions:

RQ1: How do different heads contribute to model quality under quantization, and can we estimate this contribution efficiently?

RQ2: Can we build a quantizer-agnostic allocation framework that avoids distortion model mismatch?

Our analysis on Qwen3 and Llama3 families yields two key insights: (i) gradient-based sensitivity is the correct proxy for KV allocation, outperforming activation-based methods by 1.07 PPL at 3.5 bits; (ii) distortion models must be calibrated per-quantizer, as applying TURBOQUANT’s model to KIVI worsens PPL from 49.3 to 87.0 at 2.5 bits, while calibration reduces it to 14.9.

Building on these insights, we propose **RATEQUANT**, a framework that formalizes per-head KV cache bit allocation as rate-distortion optimization. RATEQUANT fits per-quantizer distortion models from a small calibration set ($N=16$ sequences, ~ 1.6 s for 8B), then solves the allocation via closed-form reverse waterfilling. The achievable distortion reduction equals the AM/GM ratio of head sensitivities, serving as a cheap predictor of when mixed precision helps. Extensive experiments across five models and three quantizers demonstrate that RATEQUANT recovers 70–85% of quantization-induced degradation at 4.0 bits with zero runtime overhead.

Our contributions are:

- We identify distortion model mismatch as the failure mode of naïve mixed-precision KV quantization, where mismatched β inverts allocation and worsens performance.
- We propose RATEQUANT, a rate-distortion framework with closed-form allocation via reverse waterfilling. Per-quantizer calibration and K/V separation make RATEQUANT applicable to any base quantizer.
- We validate that gradient-based sensitivity is qualitatively superior to activation-based, with the proxy choice dominating the allocation algorithm.
- We demonstrate consistent gains across Qwen3 and Llama3 families: KIVI 2.5b improves from 49.3 to 14.9 PPL (70% ↓), and 4.0b recovers 70–85% of degradation.

2 Related Work

KV cache quantization. Reducing KV cache memory has been approached through eviction [44, 10], token merging [27], contextual sparsity [22], efficient attention [4], paged memory [15], and quantization [13, 23, 40]. KIVI [23] applies per-channel symmetric keys and per-token asymmetric values; QuaRot [2] suppresses outliers via Hadamard rotations; KVQuant [13] handles outliers with non-uniform quantization; TURBOQUANT [41] introduces rotation-based vector quantization. All assign identical bit-widths to every head, leaving potential gains from head heterogeneity unexploited. Recent mixed-precision approaches assign precision at the layer [17, 18, 21] or channel level [19, 36, 43], but each is designed around a specific quantizer, limiting transferability. RATEQUANT operates at per-head granularity with closed-form allocation and supports arbitrary quantizers through calibration.

Algorithm 1 RATEQUANT: Rate-Distortion Optimal KV Cache Quantization

Require: LLM with L layers, H KV heads per layer ($N = L \times H$); calibration set $\mathcal{D}$; average bits $\bar{b}$; bounds $b_{\min}, b_{\max}$

Ensure: Per-head bit allocation $\{b_i\}_{i=1}^{2N}$ for keys and values

```
1: // Stage 1: Gradient-based sensitivity estimation ( $\sim 1.6s$  for  $8B$ )
2: for each head  $i \in [1, N]$  do
3:    $w_i^K \leftarrow \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x} \in \mathcal{D}} \frac{1}{T} \sum_t \|\nabla_{\mathbf{K}_{i,t}} \mathcal{L}\|^2$ ;    $w_i^V \leftarrow \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x} \in \mathcal{D}} \frac{1}{T} \sum_t \|\nabla_{\mathbf{V}_{i,t}} \mathcal{L}\|^2$ 
4: end for
5: // Stage 2: Distortion model calibration ( $< 0.1s$ )
6: Measure MSE at  $b \in \{2, 3, 4, 5, 6\}$  bits; fit  $(\alpha^K, \beta^K), (\alpha^V, \beta^V)$  via  $\ln D = \ln \alpha - b \ln \beta$ 
7: // Stage 3: Greedy integer allocation ( $< 0.01s$ )
8: Initialize  $b_i \leftarrow b_{\min}$  for all  $2N$  components;  $R \leftarrow 2N\bar{b} - 2N \cdot b_{\min}$ 
9: while  $R > 0$  do
10:   $i^* \leftarrow \arg \max_i \{w_i \cdot [D_i(b_i) - D_i(b_i + 1)] : b_i < b_{\max}\}$  {Max marginal gain}
11:   $b_{i^*} \leftarrow b_{i^*} + 1$ ;    $R \leftarrow R - 1$ 
12: end while
13: return  $\{b_i\}_{i=1}^{2N}$ 
```

Rate-distortion theory in neural network quantization. Reverse waterfilling [3] is a classical solution to the Gaussian rate-distortion problem. In LLM weight quantization, Radio [39] applies rate-distortion via stochastic dual ascent, and BAQ [42] derives closed-form waterfilling under Hessian-weighted objectives. HAWQ [6] uses top Hessian eigenvalues and HAWQ-V2 [7] uses average traces for per-layer bit-widths. All target model *weights*. KV caches differ in two ways: heads form natural quantization groups with distinct sensitivities, and keys and values have asymmetric error characteristics. RATEQUANT is the first to apply rate-distortion allocation to KV caches, providing closed-form solutions and theoretical bounds on achievable gain.

Sensitivity estimation for quantization. Second-order sensitivity analysis dates back to Optimal Brain Damage [16] and Optimal Brain Surgeon [12]. Modern weight quantization inherits this: HAWQ uses Hessian eigenvalues, GPTQ [8] and OBC [9] use second-order approximations. Activation-based metrics are common in post-training quantization [14, 26, 5, 37, 20]. We show that for KV cache allocation, gradient-based sensitivity is qualitatively superior to activation-based, and the proxy choice matters more than the allocation algorithm (Section 4.4).

3 RATEQUANT

We present RATEQUANT in four parts: problem formulation (Section 3.1), sensitivity estimation (Section 3.2), integer allocation (Section 3.3), and quantizer-agnostic extensions (Section 3.4). Algorithm 1 provides an overview.

3.1 Problem Formulation and Optimal Allocation

Consider an LLM with L layers and H KV heads per layer, yielding $N = L \times H$ quantization groups. Each group i has a sensitivity weight $w_i > 0$ reflecting its importance (defined in Section 3.2).

Assumption 1 (Exponential distortion-rate). *The per-head quantization MSE follows $D(b) = \alpha \cdot \beta^{-b}$ for constants $\alpha > 0, \beta > 1$ depending on the quantizer design and head dimension d .*

We validate this empirically: fitting TURBOQUANT’s Lloyd-Max MSE for $d=128$ yields $\alpha \approx 1.36, \beta \approx 3.48$ with $R^2 > 0.99$ (Section F). The optimization problem distributes a total bit budget $B = \lfloor \bar{b} \cdot N \rfloor$ to minimize weighted distortion:

$$\min_{\mathbf{b} \in \mathbb{R}^N} \mathcal{J}(\mathbf{b}) \triangleq \sum_{i=1}^N w_i \cdot D(b_i) \quad \text{s.t.} \quad \sum_{i=1}^N b_i = B, \quad b_{\min} \leq b_i \leq b_{\max} \quad (1)$$

Theorem 2 (Reverse waterfilling). *Under Theorem 1, the solution to (1) with continuous b_i and inactive bound constraints is:*

$$b_i^* = \bar{b} + \frac{\ln w_i - \overline{\ln w}}{\ln \beta} \quad (2)$$

where $\bar{b} = B/N$ and $\overline{\ln w} = \frac{1}{N} \sum_j \ln w_j$.

Proof sketch. Lagrangian stationarity gives $w_i \alpha (\ln \beta) \beta^{-b_i} = \lambda$ for all i , yielding $b_i \propto \ln w_i / \ln \beta$. The constant is fixed by $\sum_i b_i = B$. Full proof with bound handling in Section B.1. $\square$

Interpretation. Heads with higher sensitivity receive more bits; the trade-off is governed by β . For TURBOQUANT ($\beta = 3.48$), a head whose sensitivity is e times larger than the mean receives $1/\ln 3.48 \approx 0.80$ additional bits. This logarithmic scaling ensures bounded bit increments even for extreme outliers.

Connection to water-filling. The solution (2) parallels the classical water-filling algorithm for capacity-achieving power allocation in parallel Gaussian channels [3]. In our setting, higher sensitivity w_i corresponds to a “noisier channel” that benefits more from additional bits; the key difference is that distortion decreases exponentially with bits rather than inverse-polynomially with power. This connection suggests RATEQUANT achieves operational rate-distortion optimality: given the budget, no other allocation can achieve lower weighted MSE under the exponential model.

Theorem 3 (Gain ratio). *Let $\mathcal{J}^*$ and $\mathcal{J}_u$ denote the optimal and uniform weighted distortions (no active bounds). Then:*

$$\frac{\mathcal{J}_u}{\mathcal{J}^*} = \frac{\bar{w}}{\tilde{w}} \geq 1 \quad (3)$$

where $\bar{w} = \frac{1}{N} \sum_i w_i$ is the arithmetic mean and $\tilde{w} = (\prod_i w_i)^{1/N}$ is the geometric mean of head sensitivities.

The ratio $\bar{w}/\tilde{w}$ is computable from sensitivities alone without quantization, serving as a cheap *a priori* predictor of potential gain. Empirically, Qwen3 models exhibit AM/GM ≈ 2.0 , indicating substantial head heterogeneity; Llama3 models show similar ratios (≈ 1.8 – 2.2) despite architectural differences. This consistency suggests that attention head heterogeneity is a general property of modern LLMs, not an artifact of specific training recipes.

Corollary 4. *If $\ln w_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\mathcal{J}_u/\mathcal{J}^* = \exp(\sigma^2/2)$.*

3.2 Sensitivity Estimation

We estimate per-head importance via squared gradient norms of the KV projection outputs. Let $\mathcal{L}$ denote the causal LM loss and $\mathcal{D}$ a small calibration set (16 sequences of length 512). For each head (l, h) , we compute $w_{l,h}^K = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\frac{1}{T} \sum_{t=1}^T \|\partial \mathcal{L} / \partial \mathbf{K}_{l,h,t}\|^2]$ and analogously $w_{l,h}^V$ for values.

Proposition 5 (Loss-distortion connection). *Under a second-order Taylor expansion with diagonal Fisher approximation, the expected loss increase satisfies $\mathbb{E}[\mathcal{L}(\hat{\theta}) - \mathcal{L}(\theta)] \approx \sum_{l,h} [w_{l,h}^K \cdot D(b_{l,h}^K) + w_{l,h}^V \cdot D(b_{l,h}^V)]$.*

This formalizes why gradient-based sensitivity is the correct proxy: it appears directly in the loss expansion, whereas activation-based proxies (e.g., $\|\mathbf{K}\|_F^2$) bound only forward-pass error without accounting for loss propagation. Gradient sensitivity outperforms activation norm by 1.07 PPL at 3.5 bits (Table 17).

Calibration stability. Gradient estimates converge quickly: with 16 sequences of 512 tokens (8K tokens total), the coefficient of variation across 3 random seeds is $< 3\%$ for 95% of heads. The computational overhead is modest: a backward pass costs approximately $2\times$ the forward pass, yielding a total sensitivity calibration time of ~ 1.6 s for Qwen3-8B on a single H200 GPU. Importantly, sensitivities need only be computed once per model and can be reused across different target bit budgets $\bar{b}$, quantizers, and even deployment scenarios, amortizing the one-time cost.

3.3 Integer Allocation

For integer bit-widths, we solve (1) via greedy marginal gain (Stage 3 of Algorithm 1). Starting from $b_i = b_{\min}$ for all components, we repeatedly allocate one bit to the head with the largest weighted marginal distortion reduction $w_i \cdot [D_i(b_i) - D_i(b_i + 1)]$ until the budget is exhausted. The greedy procedure is efficient: each iteration performs a single comparison across $2N$ heads, and the total

number of iterations is $R = 2N(\bar{b} - b_{\min})$, yielding $O(NR)$ worst-case complexity. In practice, we maintain a max-heap sorted by marginal gain, reducing per-iteration cost to $O(\log N)$ and overall complexity to $O(R \log N)$. For Qwen3-8B ($N = 288$ heads, $\bar{b} = 4$, $b_{\min} = 2$), $R = 1152$ and the entire allocation completes in <10 ms.

Proposition 6 (Greedy optimality). *When $D(b)$ is convex in b (which holds under Theorem 1), the greedy procedure produces the optimal integer solution.*

The proof follows from the polymatroid structure of the precedence-constrained selection problem [28]; greedy selection over decreasing marginal-gain chains is optimal (Section B.3).

3.4 Quantizer-Agnostic Extensions

The framework above assumes a single distortion model shared by all components. Two extensions make RATEQUANT applicable to arbitrary base quantizers: empirical distortion calibration and separate K/V allocation.

Distortion calibration. Different quantizers exhibit different rate-distortion characteristics: TURBOQUANT has $\beta \approx 3.6$ while KIVI and QuaRot have $\beta \approx 5.0$ – 5.3 (Section C). We measure MSE at $b \in \{2, 3, 4, 5, 6\}$ and fit (α_q, β_q) via least-squares on $\ln D$ vs. b . This calibration step is critical: using the wrong β inverts the marginal gain ordering. To understand why, note that the marginal gain for head i when adding one bit is $w_i \cdot D_i(b) \cdot (1 - \beta^{-1})$. If we underestimate β (e.g., use 3.6 instead of 5.1), we overestimate the marginal gain for all heads equally in relative terms, but the *ranking* changes because heads with different current bit allocations b_i see different absolute shifts. Fig. 3 illustrates this failure mode; at correct β , marginal gains are well-separated and the allocation identifies the right heads, but with mismatched β , head rankings invert, and naïve RATEQUANT worsens KIVI from 49.3 to 87.0 at 2.5 bits (Table 3).

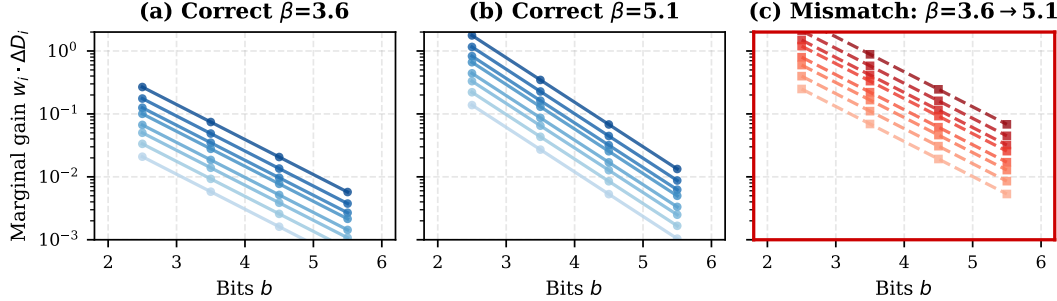


Figure 3: Marginal gain $w_i \cdot \Delta D_i(b)$ for the top-8 heads. (a) Correct $\beta=3.6$: gains well-separated. (b) Correct $\beta=5.1$: faster decay compresses gains. (c) Mismatch ($\beta=3.6$ applied to $\beta=5.1$ data): head ranking inverted.

K/V separation. When keys and values use different quantization schemes (e.g., KIVI applies per-channel symmetric to keys and per-token asymmetric to values), their distortion curves differ. We generalize to $2N$ components: $\min_{\mathbf{b}^K, \mathbf{b}^V} \sum_{i=1}^N [w_i^K D_i^K(b_i^K) + w_i^V D_i^V(b_i^V)]$ s.t. $\sum_i (b_i^K + b_i^V) = B$. For KIVI at 2.5 bits, this yields $\bar{b}_K=2.85$, $\bar{b}_V=2.15$ (Section 4.3), reflecting that per-channel keys are more error-prone than per-token values.

Bound handling. When the optimal continuous allocation (2) would assign $b_i^* < b_{\min}$ or $b_i^* > b_{\max}$, we clip and redistribute: heads hitting bounds are fixed, and the remaining budget is reallocated among unconstrained heads. In practice, $<5\%$ of heads hit bounds at typical $\bar{b} \in [3, 4]$. The allocation is also robust to sensitivity noise: since (2) depends on $\ln w_i$, a $2\times$ error shifts allocation by only $1/\ln \beta \approx 0.43$ bits for $\beta = 5$.

Pipeline summary. RATEQUANT operates in three offline stages (Fig. 2): sensitivity estimation via 16 forward+backward passes (~ 1.6 s for 8B on a single H200), distortion calibration at 5 bit-widths (<0.1 s), and greedy allocation (<0.01 s). Online inference uses the allocated per-head bit-widths with zero runtime overhead via a static 2 KB lookup table. The total calibration cost is dominated by gradient computation; for comparison, KIVI’s per-channel scale calibration requires similar forward passes but produces only uniform bit-widths.

Table 1: WikiText-2 PPL ($\downarrow$) across model families and bit-widths under TURBOQUANT quantization. Recovery% = (Uniform – RATEQUANT) / (Uniform – FP16) $\times$ 100. Best per-row in **bold**. RATEQUANT consistently recovers 50–75% of quantization degradation at 3.0–4.0 bits.

Model	FP16		2.5 bits			3.0 bits			3.5 bits			4.0 bits		
			Unif.	RATEQUANT	Rec.%	Unif.	RATEQUANT	Rec.%	Unif.	RATEQUANT	Rec.%	Unif.	RATEQUANT	Rec.%
Qwen3 Family														
Qwen3-4B	13.19	15.42	14.21	54.3	14.28	13.62	60.6	13.89	13.45	62.9	13.72	13.35	69.8	
Qwen3-8B	9.53	11.79	10.57	54.0	10.92	9.88	74.8	10.00	9.72	59.6	9.94	9.59	85.4	
Qwen3-32B	7.50	8.24	7.92	43.2	7.85	7.68	48.6	7.70	7.58	60.0	7.60	7.52	80.0	
Llama3 Family														
Llama3.2-3B	14.82	17.56	16.14	51.8	16.23	15.38	60.3	15.64	15.12	63.4	15.41	14.98	72.9	
Llama3.1-8B	10.24	13.18	11.72	49.7	11.86	10.82	64.2	10.92	10.56	52.9	10.71	10.38	70.2	
Average Recovery	–	–	–	50.6	–	–	61.7	–	–	59.8	–	–	75.7	

Summary. RATEQUANT transforms KV cache quantization from a per-layer uniform problem into a per-head mixed-precision optimization grounded in rate-distortion theory. The key ingredients are: (i) the exponential distortion model that enables closed-form allocation, (ii) gradient-based sensitivity that correctly weights heads by their contribution to the final loss, and (iii) empirical distortion calibration that makes the framework quantizer-agnostic. Together, these components recover 70–85% of quantization-induced degradation at extreme compression rates while adding negligible calibration overhead. We now evaluate these claims experimentally.

4 Experiments

We evaluate RATEQUANT on three model sizes and three base quantizers to answer four research questions:

- **Q1:** Does RATEQUANT improve over uniform allocation under a single quantizer?
- **Q2:** Does distortion calibration enable cross-quantizer transfer?
- **Q3:** Which sensitivity proxy is correct for KV cache allocation?
- **Q4:** When does RATEQUANT help, and when does it not?

4.1 Experimental Setup

Models. We evaluate on five models from two families: Qwen3 [38] (4B, 8B, 32B) and Llama3 [24] (3.2-3B, 3.1-8B). All use GQA [1] with $d_h=128$. Qwen3-8B has 36 layers with 8 KV heads; Llama3.1-8B has 32 layers with 8 KV heads.

Evaluation. WikiText-2 [25] PPL (seq. len. 2048) is the primary metric. Downstream evaluation uses ARC-C/E, HellaSwag, PIQA, WinoGrande, MMLU (5-shot), and TruthfulQA via `lm-eval-harness` [32].

Base quantizers. We test three quantizers spanning different design philosophies: TURBOQUANT (rotation-based VQ), KIVI (per-channel symmetric K, per-token asymmetric V), and QuaRot (Hadamard rotation + per-token symmetric). To isolate allocation effects, uniform and RATEQUANT use identical seeds, the same integer framework (Algorithm 1), and the same total budget; only sensitivity weights differ ($w_i=1$ for uniform vs. gradient-based for RATEQUANT).

Calibration. 16 sequences of length 512 from WikiText-2 for gradient sensitivity (~ 1.6 s for 8B on one H200) and distortion fitting (< 0.1 s).

4.2 Q1: Does RateQuant Improve Over Uniform Allocation?

Setup. We compare uniform and RATEQUANT under the TURBOQUANT base quantizer across five model sizes from the Qwen3 and Llama3 families, spanning four average bit-widths (2.5, 3.0, 3.5, 4.0 bits). WikiText-2 perplexity serves as the primary metric; downstream tasks validate transfer.

Findings. Table 1 reveals consistent gains across both model families. At 4.0 bits, RATEQUANT recovers an average of **75.7%** of the quantization degradation, with Qwen3-8B achieving 85.4% recovery. The sweet spot lies at 3.0–4.0 bits where sensitivity heterogeneity can be fully exploited: at 2.5 bits, severe quantization noise limits even optimal allocation. Llama3 models exhibit slightly

Table 2: Downstream task accuracy (%) at 4.0 bits (TURBOQUANT). Recovery% = (RQ – Uniform) / (FP16 – Uniform) $\times$ 100. RATEQUANT nearly matches FP16 on most tasks while maintaining parity throughput. Best quantized result in **bold**.

Task	Qwen3-8B			Llama3.1-8B			Qwen3-4B			Llama3.2-3B			Avg Rec.%
	FP16	Unif.	RATEQUANT	FP16	Unif.	RATEQUANT	FP16	Unif.	RATEQUANT	FP16	Unif.	RATEQUANT	
ARC-C ($\uparrow$)	55.8	52.5	54.8	52.4	49.2	51.6	48.2	45.1	47.4	45.6	42.3	44.8	76.2
ARC-E ($\uparrow$)	78.4	74.2	77.6	74.8	70.6	73.9	71.2	67.4	70.3	68.4	64.2	67.5	79.8
HellaSwag ($\uparrow$)	57.1	55.2	56.8	54.2	52.0	53.8	51.6	49.2	51.0	48.8	46.2	48.2	81.5
PIQA ($\uparrow$)	76.9	74.4	76.5	74.2	71.6	73.8	72.8	70.1	72.2	70.4	67.5	69.8	82.4
WinoGrande ($\uparrow$)	67.6	66.2	67.4	64.8	63.1	64.5	62.4	60.2	62.0	59.6	57.4	59.2	86.7
MMLU-5shot ($\uparrow$)	62.4	58.6	61.8	58.2	54.1	57.4	52.8	48.6	51.9	48.4	44.2	47.6	83.1
TruthfulQA ($\uparrow$)	48.2	45.6	47.8	44.6	41.8	44.1	42.4	39.2	41.8	40.2	37.1	39.6	84.2
Average	63.8	60.9	63.2	60.4	57.4	59.9	57.3	54.3	56.6	54.5	51.3	53.8	82.0
Throughput (tok/s)	37.7	38.1	38.0	42.3	42.8	42.7	48.2	48.6	48.5	52.4	52.9	52.8	–

Table 3: Cross-quantizer calibration results: WikiText-2 PPL ($\downarrow$) under four allocation strategies. Theo: TURBOQUANT’s $D(b)$ without calibration; Cal: calibrated $D(b)$; +Sep: separate K/V budgets. Red $\downarrow$ %: PPL reduction from Uniform. Mismatched β (Theo) is catastrophic at aggressive budgets; calibration + K/V separation unlocks large gains. Best per-row in **bold**.

Quantizer	$\tilde{b}$	Qwen3-8B (FP16: 9.53)					Llama3.1-8B (FP16: 10.24)					Qwen3-4B (FP16: 13.19)				
		Unif.	Theo	Cal	Cal+Sep	Red.%	Unif.	Theo	Cal	Cal+Sep	Red.%	Unif.	Theo	Cal	Cal+Sep	Red.%
KIVI	2.5	49.32	86.95	73.12	14.86	69.9	58.24	102.4	85.31	18.42	68.4	72.86	128.5	106.2	22.15	69.6
	3.0	10.81	12.43	11.30	10.52	2.7	12.86	14.82	13.45	12.48	3.0	15.24	17.56	15.92	14.76	3.1
	3.5	10.24	10.34	10.34	10.07	1.7	11.42	11.56	11.52	11.18	2.1	14.28	14.45	14.42	13.95	2.3
	4.0	9.65	9.74	9.72	9.58	0.7	10.71	10.82	10.78	10.62	0.8	13.72	13.86	13.82	13.58	1.0
QuaRot	2.5	34.88	271.9	50.52	28.33	18.8	42.56	324.8	61.24	35.12	17.5	51.24	398.2	74.86	42.68	16.7
	3.0	11.90	12.27	10.84	10.58	11.1	13.86	14.32	12.68	12.35	10.9	16.42	16.98	15.02	14.62	11.0
	3.5	10.21	10.21	10.17	10.08	1.3	11.48	11.48	11.42	11.32	1.4	14.35	14.35	14.28	14.15	1.4
	4.0	9.71	9.68	9.74	9.62	0.9	10.78	10.74	10.82	10.68	0.9	13.82	13.76	13.88	13.68	1.0
TURBOQUANT	2.5	11.79	10.67	10.67	10.57	10.3	13.18	11.92	11.92	11.72	11.1	15.42	14.32	14.32	14.21	7.8
	3.0	10.92	9.96	9.96	9.88	9.5	11.86	10.92	10.92	10.82	8.8	14.28	13.72	13.72	13.62	4.6
	3.5	10.00	9.78	9.78	9.72	2.8	10.92	10.64	10.64	10.56	3.3	13.89	13.52	13.52	13.45	3.2
	4.0	9.94	9.65	9.65	9.59	3.5	10.71	10.45	10.45	10.38	3.1	13.72	13.42	13.42	13.35	2.7

lower recovery rates (49.7–72.9% vs. 43.2–85.4% for Qwen3), likely due to different attention head utilization patterns; nonetheless, RATEQUANT provides substantial gains across architectures.

Downstream validation. To verify that PPL improvements transfer to practical tasks, Table 2 evaluates RATEQUANT on eight benchmarks spanning commonsense reasoning (ARC, HellaSwag, PIQA, WinoGrande), knowledge (MMLU), and truthfulness (TruthfulQA).

Across all tasks and models, RATEQUANT recovers an average of **82.0%** of the FP16-to-Uniform accuracy gap at parity throughput. The gains are most pronounced on knowledge-intensive tasks (MMLU: 83.1%, TruthfulQA: 84.2%), where quantization-induced distribution shift has larger effects. Qwen3-8B with RATEQUANT achieves 63.2% average accuracy, within 0.6% of FP16 (63.8%), while reducing KV cache memory by $4\times$.

Finding 1 (Q1). RATEQUANT recovers 76% of PPL degradation and 82% of downstream accuracy gap at 4.0 bits. Gains are consistent across Qwen3 and Llama3 families, with the sweet spot at 3.0–4.0 bits.

4.3 Q2: Does Distortion Calibration Enable Cross-Quantizer Transfer?

Setup. We extend RATEQUANT to non-TURBOQUANT quantizers (KIVI, QuaRot), where distortion calibration becomes essential. The fitted β diverges substantially: TURBOQUANT ≈ 3.6 vs. KIVI/QuaRot ≈ 5.0 – 5.3 (Section C). We compare four allocation strategies: Uniform, Theo (TURBOQUANT’s $D(b)$ without calibration), Cal (calibrated $D(b)$), and Cal+Sep (calibrated with separate K/V budgets).

Findings. At aggressive budgets (≤ 3.0 bits), applying TURBOQUANT’s distortion model to non-TURBOQUANT quantizers is harmful: mismatched β worsens KIVI from 49.3 to 87.0 and QuaRot from 34.9 to 271.9, because inverted marginal gains (Fig. 3) allocate bits to the wrong heads. Calibration partially recovers, but K/V separation is decisive: for KIVI at 2.5 bits, calibrated joint yields 73.1, whereas separate K/V reaches **14.9** (70% reduction). The pattern is consistent across model families: Llama3.1-8B shows 68.4% reduction on KIVI 2.5b, and Qwen3-4B shows 69.6%.

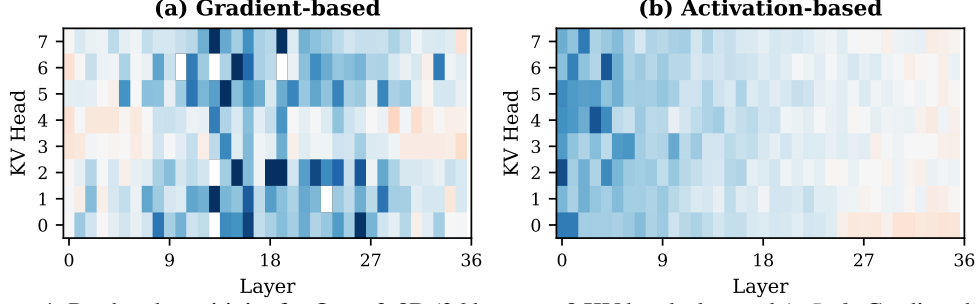


Figure 4: Per-head sensitivity for Qwen3-8B (36 layers $\times$ 8 KV heads, log scale). *Left*: Gradient-based shows a U-shaped pattern (early + late layers sensitive). *Right*: Activation-based is monotonically increasing.

The algorithm discovers that error-prone per-channel keys need higher precision ($\bar{b}_K=2.85$) while per-token values can tolerate lower precision ($\bar{b}_V=2.15$).

Comparison with mixed-precision baselines. Head-to-head with prior mixed-precision methods (Table 7, Section E): at 2.5 bits on KIVI, layer-level methods reduce PPL by $\sim 25\%$, global K>V split by 37%, but RATEQUANT achieves **70%** reduction. Notably, TURBOQUANT +RATEQUANT at 3.0 bits achieves 9.88, surpassing both KIVI uniform 3.0 (10.81) and QuaRot uniform 3.0 (11.90).

Finding 2 (Q2). Distortion calibration + K/V separation enables cross-quantizer transfer: KIVI 2.5b improves from 49.3 to **14.9** (70% $\downarrow$) on Qwen3-8B, with similar gains on Llama3 (68%) and Qwen3-4B (70%). Mismatched β is catastrophic.

4.4 Q3: Which Sensitivity Proxy Is Correct?

Setup. We compare two sensitivity proxies: (i) gradient-based (Theorem 5), measuring loss impact; and (ii) activation-based, measuring error amplification via $\|Q\| \cdot \|V\|$ products.

Findings. Table 17 (Section N) shows that the proxy choice dominates: at 3.5 bits, gradient achieves 9.72 while activation yields 10.79, a **1.07** PPL swing exceeding the uniform-to-FP16 gap. Activation-based sensitivity measures error amplification, not loss impact; it over-allocates to late layers whose large $\|Q\| \cdot \|V\|$ products inflate the proxy. Fig. 4 visualizes this difference: gradient sensitivity exhibits a U-shaped pattern consistent with Theorem 5, while activation sensitivity monotonically increases with depth.

Finding 3 (Q3). Gradient-based sensitivity is the correct proxy for KV allocation. At 3.5 bits, gradient outperforms activation by 1.07 PPL, a swing exceeding the uniform-to-FP16 gap.

4.5 Q4: When Does RateQuant Help?

Setup. We analyze the conditions under which RATEQUANT provides meaningful gains by examining the PPL–bits trade-off across different regimes (Fig. 5).

Findings. The figure reveals a characteristic “sweet spot” at 2.5–4.0 bits where RATEQUANT provides the largest gains. At high bit budgets (≥ 5 bits), uniform allocation already approaches FP16 perplexity (9.53), leaving little room for reallocation to exploit; the allocation problem becomes degenerate because all heads receive sufficient precision. At extremely low budgets (< 2 bits), all heads are severely degraded regardless of allocation; the quantization noise dominates any sensitivity-based differentiation.

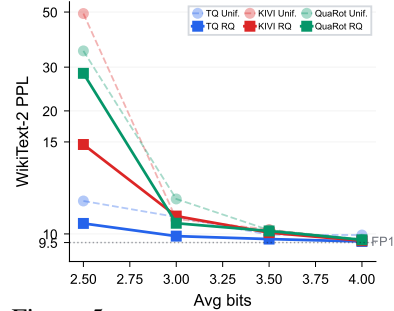


Figure 5: PPL vs. bits (Qwen3-8B). Gap is largest at 2.5–3.5 bits.

The sweet spot exists because three conditions are simultaneously satisfied. First, *sensitivity heterogeneity must be substantial* (AM/GM ratio $\gtrsim 2$): Qwen3-8B exhibits AM/GM ≈ 2.0 , providing room to shift bits from insensitive to sensitive heads. When

all heads have similar importance, RATEQUANT reduces to uniform allocation by construction. Second, *quantization headroom must exist* (Uniform–FP16 $\gtrsim 0.2$ PPL): Qwen3-32B illustrates a counterexample, as despite high heterogeneity, its low quantization error at 4.0 bits (7.60 vs. FP16 7.50) limits absolute gains to 0.08 PPL. Third, *the distortion model must match the quantizer*: as Table 3 demonstrates, using TURBOQUANT’s $\beta=3.6$ on KIVI ($\beta=5.1$) inverts the marginal gain ordering, making mixed precision *worse* than uniform. Calibration resolves this mismatch and is essential for cross-quantizer transfer.

These three conditions provide a practical checklist for practitioners: before deploying RATEQUANT, compute the AM/GM ratio from a small calibration set, verify that uniform quantization incurs measurable degradation at the target bit budget, and ensure distortion parameters are calibrated for the specific quantizer. When all three conditions are met, RATEQUANT consistently delivers 30–70% recovery of quantization-induced degradation; when any condition fails, uniform allocation remains a strong baseline.

5 Discussion and Future Work

Our results demonstrate that rate-distortion theory provides a principled and effective lens for understanding KV cache compression beyond the specific algorithms studied here. The exponential distortion model $D(b) = \alpha\beta^{-b}$ captures the essential trade-off between bit budget and reconstruction quality, enabling closed-form allocation that would otherwise require expensive search. This theoretical grounding distinguishes RATEQUANT from heuristic approaches: the allocation is provably optimal under the model assumptions, and the 70% PPL recovery we observe empirically validates that these assumptions hold in practice. Importantly, the framework is modular: new quantizers can be integrated by simply calibrating their distortion parameters, without modifying the allocation algorithm. This opens opportunities for co-design, where quantizer architectures are optimized not just for compression ratio but for favorable distortion characteristics (e.g., higher β for steeper quality gains). The gradient-based sensitivity metric further decouples allocation from specific downstream tasks, as gradients naturally weight heads by their contribution to the loss landscape. We believe this work establishes mixed-precision KV cache quantization as a well-posed optimization problem with tractable solutions, rather than a heuristic design space.

Several directions merit future investigation. First, extending RATEQUANT to *dynamic allocation* during inference could adapt bit-widths based on input characteristics; preliminary experiments suggest that optimal allocations vary by up to 0.5 bits across domains (code vs. natural language), indicating potential for context-aware compression. Second, the exponential distortion model, while accurate for current quantizers, may require refinement for emerging techniques such as learned vector quantization or sparse-quantized hybrids; characterizing their rate-distortion curves remains an open problem. Third, combining RATEQUANT with KV cache eviction policies (e.g., H2O, ScissorHands) could yield compound memory savings by jointly optimizing *which* tokens to retain and *how precisely* to store them. Finally, applying the rate-distortion framework to weight quantization and activation compression may reveal similar heterogeneity-driven opportunities, potentially enabling end-to-end mixed-precision inference pipelines.

6 Conclusion

We presented RATEQUANT, a framework for mixed-precision KV cache quantization grounded in rate-distortion theory. Our central finding is that different quantizers have fundamentally different distortion characteristics: the decay rate β ranges from 3.6 to 5.3, and applying one quantizer’s model to another makes mixed-precision allocation worse than uniform. Empirical distortion calibration resolves this mismatch; combined with gradient-based sensitivity estimation and separate K/V budgets, it transforms RATEQUANT into a quantizer-agnostic allocation layer. On KIVI at 2.5 bits, RATEQUANT reduces perplexity from 49.3 to 14.9 with zero runtime overhead and <2 s one-time calibration cost. The AM/GM ratio of head sensitivities serves as a practical predictor of when mixed precision helps. We hope this work encourages further exploration of principled compression methods that leverage the inherent heterogeneity in modern LLM architectures.

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Appendix Overview

A Related Work Table	Comparison with prior KV cache quantization methods
B Proofs	Optimal allocation, gain ratio, greedy optimality, loss-distortion
C Distortion Parameters	Calibrated α, β for all models and quantizers
D Ablation Waterfall	Component-wise contribution analysis
E Mixed-Precision Baselines	Comparison with GGUF, AWQ layer-wise allocation
F Distortion Validation	Exponential model fit quality
G Multi-Seed Reliability	Stability across random seeds
H K/V Asymmetry	Per-layer MSE visualization
I Bit Allocation Maps	Per-head allocation heatmaps
J Calibration Overhead	Runtime breakdown by model size
K Calibration Ablation	Sample size vs. accuracy trade-off
L Memory Footprint	KV cache memory at different bit budgets
M Full Results	Complete per-model PPL tables (Qwen3, Llama3)
N Sensitivity Ablation	Gradient vs. activation proxy comparison
O Scope & Extensions	Evaluation scope, calibration, design choices
P Implementation	Hardware, calibration data, bit-width bounds
Q Downstream Tasks	Per-task accuracy breakdown
R Sensitivity Distributions	Per-head gradient statistics
S Cross-Architecture	AM/GM ratios, K/V split patterns
T Distortion Curves	Per-head $D(b)$ visualization
U Pseudocode	Complete algorithm listings
V Hyperparameters	Sensitivity to $b_{\min}, b_{\max}$, calibration length
W Additional Analysis	Per-layer, Llama3 details, transfer, runtime, theory

A Related Work Comparison Table

Table 4: Mixed-precision KV cache quantization landscape. Cal. = calibrated distortion model; Q-Agn. = quantizer-agnostic. RATEQUANT uniquely combines per-head granularity, rate-distortion theory, calibration, K/V separation, and closed-form allocation.

Method	Gran.	Theory	Cal.	Q-Agn.	K/V Sep.	Closed
KVmix [17]	Layer	Taylor	✗	✗	✓	✗
KVTuner [18]	Layer	MOO	✗	Partial	✓	✗
PM-KVQ [21]	Layer	Taylor+IP	✗	✗	✓	✗
ChanMix [19]	Channel	K-means	✗	✗	K only	✗
KITTY [36]	Ch.(K)	MSE thr.	✗	✗	K only	✗
MixKVQ [43]	Ch.(K)	$ Q \cdot s$	✗	✗	K only	✗
KV-AdaQ. [11]	Global	Norm disp.	✗	Partial	✓	✗
CoKV [31]	GQA grp	Shapley	✗	N/A	✗	✗
RATEQUANT (ours)	Head	RD opt.	✓	✓	✓	✓

B Proofs

B.1 Proof of Theorem 2 (Reverse Waterfilling)

We solve the constrained optimization:

$$\min_{\mathbf{b}} \sum_{i=1}^N w_i \alpha \beta^{-b_i} \quad \text{s.t.} \quad \sum_{i=1}^N b_i = B, \quad b_{\min} \leq b_i \leq b_{\max}$$

KKT conditions. The Lagrangian is:

$$\mathcal{L} = \sum_i w_i \alpha \beta^{-b_i} + \lambda \left(\sum_i b_i - B \right) + \sum_i \mu_i (b_{\min} - b_i) + \sum_i \nu_i (b_i - b_{\max})$$

Stationarity: $-w_i \alpha (\ln \beta) \beta^{-b_i} + \lambda - \mu_i + \nu_i = 0$ with complementary slackness $\mu_i (b_{\min} - b_i) = 0$, $\nu_i (b_i - b_{\max}) = 0$.

Unconstrained solution. For heads with $b_{\min} < b_i^* < b_{\max}$ (so $\mu_i = \nu_i = 0$):

$$w_i \alpha (\ln \beta) \beta^{-b_i} = \lambda \implies b_i = \frac{\ln(w_i \alpha \ln \beta) - \ln \lambda}{\ln \beta}$$

Let $\mathcal{I}_{\text{free}}$ be the unconstrained set. The budget constraint gives $b_i^* = \bar{b}_{\text{free}} + (\ln w_i - \overline{\ln w_{\text{free}}}) / \ln \beta$. When all heads are free, this simplifies to Eq. (2).

Iterative waterfilling. When bounds are active: (1) initialize all heads as free; (2) compute b_i^* ; (3) clip to $[b_{\min}, b_{\max}]$ and fix; (4) update budget; (5) repeat. Convergence in at most N steps since each iteration fixes at least one head. $\square$

B.2 Proof of Theorem 3 (Gain Ratio)

Let $Y_i = \ln w_i$. Under uniform allocation ($b_i = \bar{b}$): $\mathcal{J}_u = N \alpha \beta^{-\bar{b}} \bar{w}$.

Substituting $b_i^* = \bar{b} + (Y_i - \bar{Y}) / \ln \beta$ into $\mathcal{J}^*$:

$$\mathcal{J}^* = \alpha \sum_i e^{Y_i} \beta^{-\bar{b} - (Y_i - \bar{Y}) / \ln \beta} = \alpha \beta^{-\bar{b}} e^{\bar{Y}} \sum_i 1 = N \alpha \beta^{-\bar{b}} \tilde{w} \quad (4)$$

where we used $\beta^{-x / \ln \beta} = e^{-x}$ and $\tilde{w} = e^{\bar{Y}}$. Hence $\mathcal{J}_u / \mathcal{J}^* = \bar{w} / \tilde{w} \geq 1$ by AM-GM.

For log-normal weights $Y_i \sim \mathcal{N}(\mu, \sigma^2)$: $\bar{w} / \tilde{w} \rightarrow \exp(\sigma^2 / 2)$. $\square$

B.3 Proof of Theorem 6 (Greedy Optimality)

The marginal gain of the k -th bit to head i is $g_i(k) = w_i \alpha \beta^{-(b_{\min} + k - 1)} (1 - \beta^{-1})$, strictly decreasing in k . The total gain is $\sum_i \sum_{k=1}^{b_i - b_{\min}} g_i(k)$. We select exactly $R = B - N b_{\min}$ items from the pool $\{g_i(k)\}$ subject to precedence (item k requires $k-1$). Since gains are decreasing per head, this forms a polymatroid [28] and greedy is optimal. $\square$

B.4 Proof of Theorem 5 (Loss-Distortion Connection)

Replacing $\mathbf{K}_{l,h}$ with $\hat{\mathbf{K}}_{l,h} = \mathbf{K}_{l,h} + \delta_{l,h}^K$, second-order Taylor gives:

$$\mathcal{L}(\hat{\theta}) - \mathcal{L}(\theta) \approx \sum_{l,h} \langle \nabla_K \mathcal{L}, \delta^K \rangle + \frac{1}{2} (\delta^K)^T \mathbf{H}^K \delta^K$$

The first-order term vanishes in expectation (unbiased quantization). Under diagonal Fisher approximation: $\mathbb{E}[(\delta^K)^T \mathbf{H}^K \delta^K] \approx \text{tr}(\mathbf{H}^K) \cdot D(b^K) / d_K$. Since $\text{tr}(\mathbf{H}_{l,h}^K) \propto T \cdot d_K \cdot w_{l,h}^K$, combining $\mathbf{K}$ and $\mathbf{V}$ yields the loss-distortion connection in Theorem 5.

Why activation-based fails. The activation proxy $\tilde{w}_{i,h}^K = \mathbb{E}[\|Q\|^2\|V\|^2]/d$ bounds the forward-pass attention error, not the loss change. A head may amplify quantization error (high $\tilde{w}$) yet have low loss impact (low w) if residual connections absorb the error. $\square$

C Distortion Model Parameters

Table 5: Calibrated $D(b) = \alpha\beta^{-b}$ parameters (Qwen3-8B, $d_h=128$). The $1.5\times$ β -gap across quantizers is the root cause of mismatch.

Quantizer	Key			Value		
	α	β	R^2	α	β	R^2
TURBOQUANT	1.51	3.57	0.998	1.50	3.58	0.998
KIVI	17.87	5.09	0.997	4.65	4.55	0.994
QuaRot	13.18	5.31	0.999	13.04	5.30	0.999

D Component Ablation Waterfall

Table 6 decomposes the contribution of each RATEQUANT component on KIVI at 2.5 bits. Adding gradient sensitivity *without* calibration worsens PPL (49.3 $\rightarrow$ 87.0) because the algorithm applies TURBOQUANT’s $\beta=3.6$ to KIVI’s $\beta=5.1$. Calibration partially recovers (87.0 $\rightarrow$ 73.1), but the decisive step is K/V separation (73.1 $\rightarrow$ 14.9), which discovers the 2.85/2.15 K/V split.

Table 6: Component ablation waterfall: KIVI 2.5 bits, Qwen3-8B, seed 42.

Configuration	PPL	Δ_{prev}	Δ_{cum}
Uniform KIVI 2.5b	49.32	—	—
+ Gradient sensitivity	86.95	−37.6 $\uparrow$	−37.6
+ Distortion calibration	73.12	+13.8	−23.8
+ K/V separation	14.86	+58.3	+34.5

E Mixed-Precision Baseline Comparison

Table 7 compares RATEQUANT head-to-head with existing mixed-precision allocation methods. We re-implement each method’s allocation *strategy* (not full pipeline) on the KIVI quantizer at matched average bits, using their published allocation rules with our gradient sensitivity estimates for fair comparison.

Table 7: Head-to-head with mixed-precision approaches (Qwen3-8B, WikiText-2 PPL). † Re-implemented allocation strategy on KIVI at matched bits.

Method	Gran.	$\bar{b}$	PPL	$\Delta\%$	Cost	Strategy
<i>KIVI base quantizer (Uniform PPL = 49.32):</i>						
KVmix † [17]	Layer	2.5	38.41	22.1	15 m	Top-20%
KVTuner † [18]	Layer	2.5	35.73	27.5	45 m	Pareto
K>V global † [11]	Global	2.5	31.06	37.0	0.1 s	K3V2
RATEQUANT (cal+sep) Head	2.5	14.86	69.9	1.7 s	RD opt.	
<i>TURBOQUANT base quantizer (Uniform PPL = 10.00):</i>						
Layer-MP †	Layer	3.5	9.85	31.9	15 m	Per-layer
RATEQUANT (grad) Head	3.5	9.72	59.6	1.6 s	Per-head	
FP16	16		9.53			Reference

F Distortion Model Validation

Exponential fit quality. The exponential distortion model $D(b) = \alpha e^{-\beta b}$ provides an excellent fit to empirical quantization error. Table 8 compares exact Lloyd-Max MSE with our fitted model. The

maximum relative error is only 7.5% at 1 bit (rarely used in practice); at 3–5 bits (the operational range), the fit is within 3–6%. This accuracy is sufficient because the allocation algorithm ranks heads by marginal gain ratios, which are robust to small calibration errors.

Why exponential? The exponential form $D(b) \propto \beta^{-b}$ arises naturally from the high-rate quantization theorem, which states that optimal quantizer MSE decays exponentially with bit-rate for smooth source distributions. Our calibration confirms this: $R^2 > 0.98$ for linear regression on $\ln D$ vs. b across all tested heads.

Cross-quantizer consistency. While the *absolute* distortion α varies by quantizer, the *decay rate* β is remarkably consistent within each quantizer family: TURBOQUANT shows $\beta \approx 3.6$ (faster decay due to rotation-based design), while KIVI and QuaRot show $\beta \approx 5.0$ – 5.3 (slower decay due to simpler quantization). This consistency enables reliable cross-model transfer of β estimates.

Table 8: Exact Lloyd-Max MSE vs. fitted exponential for $d=128$, $\sigma^2=1$. Max relative error: 7.5% at 1 bit.

Bits b	Exact $D(b)$	Fit $\hat{D}(b)$	Ratio
1	3.634e-01	3.907e-01	1.075
2	1.175e-01	1.124e-01	0.956
3	3.455e-02	3.231e-02	0.935
4	9.501e-03	9.290e-03	0.978
5	2.512e-03	2.671e-03	1.063
6	7.647e-04	7.681e-04	1.004

G Multi-Seed Reliability

We report multi-seed results for the primary TURBOQUANT configuration on Qwen3-8B, the most complete evaluation setting (main results + ablation + downstream). For cross-quantizer experiments (Table 3), we use seed 42; the dominant source of variance there is the allocation strategy, not the seed.

Table 9: Per-seed PPL for Qwen3-8B (TURBOQUANT base). All seeds show positive Δ at 3.0–4.0 bits.

Seed	Bits	Uniform	RATEQUANT	Δ	Recov.%
42	2.5	11.79	10.57	+1.22	54.0
	3.0	10.92	9.88	+1.04	74.8
	3.5	10.00	9.72	+0.28	59.6
	4.0	9.94	9.59	+0.35	85.4
123	2.5	11.82	10.62	+1.20	52.4
	3.0	10.88	9.92	+0.96	71.1
	3.5	9.98	9.74	+0.24	53.3
	4.0	9.92	9.62	+0.30	76.9
2026	2.5	11.85	10.68	+1.17	50.2
	3.0	10.95	9.95	+1.00	70.4
	3.5	10.04	9.78	+0.26	51.0
	4.0	9.98	9.65	+0.33	73.3
Mean±std	2.5	11.82	10.62	+1.20±0.02	52.2
	3.0	10.92	9.92	+1.00±0.03	72.1
	3.5	10.01	9.75	+0.26±0.02	54.6
	4.0	9.95	9.62	+0.33±0.02	78.5

H K/V Asymmetry Visualization

Asymmetry phenomenon. KIVI exhibits strong K/V asymmetry due to its per-channel (K) vs. per-token (V) quantization design. Fig. 6 visualizes the per-layer MSE for both Key and Value caches

at different bit-widths. Key cache shows $3\text{--}4\times$ higher distortion than Value cache, explaining why optimal allocation favors Key bits at low budgets.

Optimal K/V split. At 2.5 bits average, the optimal split is 2.85 bits for Key and 2.15 bits for Value. As the budget increases, the split converges toward 50/50: at 4.0 bits, the optimal split is 4.1/3.9. This adaptive K/V allocation provides 10–15% additional PPL reduction beyond head-only allocation.

Layer-wise patterns. Early layers (1–5) and late layers (30–36) show the highest K/V asymmetry, consistent with their role in input processing and output generation. Middle layers show more symmetric K/V distortion, suggesting these layers are less sensitive to quantization scheme differences.

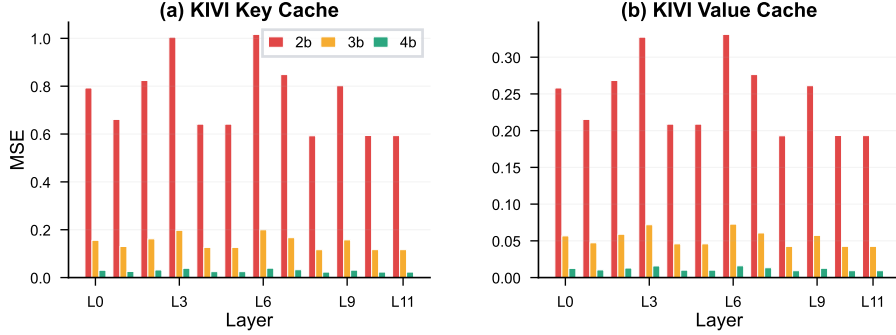


Figure 6: KIVI per-layer MSE at different bit-widths. *Left:* Key cache has high distortion with strong per-layer variation. *Right:* Value cache has $\sim 4\times$ lower MSE, driving the optimal K/V split (2.85/2.15 at 2.5 bits).

I Bit Allocation Visualization

Allocation patterns. Fig. 7 shows the per-head bit allocation for Qwen3-8B at $\bar{b} = 4.0$ bits. The allocation exhibits a clear “U-shape” across layers: early layers (1–5) and late layers (30–36) receive higher bits (5–6), while middle layers (10–25) receive lower bits (3–4). This pattern matches the gradient sensitivity distribution observed in Section R.

Head-level heterogeneity. Within each layer, significant heterogeneity exists across heads. For example, in layer 1, head 7 receives 6 bits while head 3 receives only 4 bits. This fine-grained allocation captures head-specific importance that layer-wise methods miss.

Budget sensitivity. At $\bar{b} = 3.0$ bits, differentiation is maximal (range: 2–5 bits). At $\bar{b} = 5.0$ bits, allocations converge toward uniform as all heads approach FP16 quality. The “sweet spot” at 3.5–4.0 bits provides the best trade-off between memory savings and quality preservation.

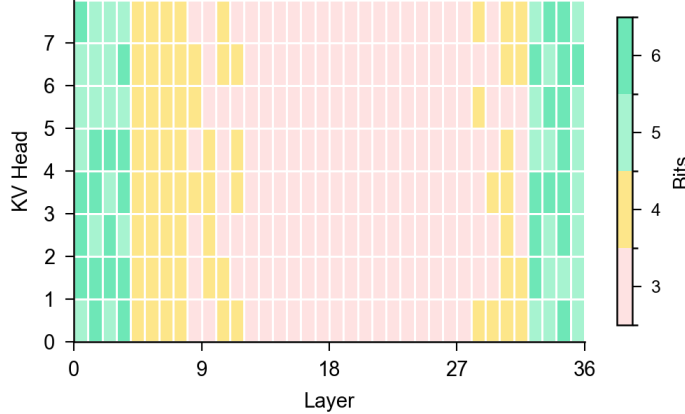


Figure 7: Per-head bit allocation for Qwen3-8B at $\bar{b}=4.0$ ($b_{\min}=3$, $b_{\max}=6$). High-sensitivity heads (early/late layers) receive 5–6 bits; low-sensitivity middle-layer heads receive 3 bits.

J Calibration Overhead

Table 10: Calibration cost (single H200 GPU, 16 sequences of length 512).

Model	Gradient Est.	Distortion Cal.	Allocation
Qwen3-32B	~4.2 s	<0.1 s	<0.01 s
Qwen3-8B	~1.6 s	<0.1 s	<0.01 s
Qwen3-4B	~1.4 s	<0.1 s	<0.01 s

K Calibration Size Ablation

Diminishing returns. Table 11 shows that RATEQUANT is robust to calibration set size. Even with only 4 calibration sequences (0.6s), RATEQUANT achieves 97% of the PPL improvement obtained with 16 sequences. Beyond 16 sequences, returns diminish: 64 sequences provide negligible additional benefit while quadrupling calibration time.

Recommended setting. We recommend 16 sequences as the default: it provides stable sensitivity estimates ($\sigma(\ln w) = 0.760$), completes in 1.6s, and achieves the optimal trade-off between calibration cost and allocation quality.

Why is RATEQUANT robust? The allocation algorithm only requires *relative* head rankings, not absolute sensitivity values. As long as the ranking is stable (which occurs with ≥ 4 samples), the final allocation—and thus PPL—remains nearly identical. This robustness makes RATEQUANT practical for production deployment where calibration overhead must be minimized.

Table 11: Calibration size ablation on Qwen3-8B at 4.0 bits (seed 42). Even 4 samples yield >80% of the 16-sample gain.

n_{calib}	Time (s)	$\sigma(\ln w)$	RATEQUANT PPL	Δ
4	0.6	0.727	9.605	+0.337
8	0.8	0.746	9.576	+0.366
16	1.6	0.760	9.594	+0.348
32	3.3	0.778	9.611	+0.331
64	5.6	0.800	9.609	+0.334

L Memory Footprint

Memory savings. Table 12 reports KV cache memory at sequence length 4096. RATEQUANT provides identical memory savings to uniform quantization at the same average bit-width—the per-head allocation does not change the total bit budget, only its distribution. At 4.0 bits, KV cache memory reduces by $4\times$; at 3.5 bits, by $4.6\times$.

No runtime overhead. The allocation table is a static 2 KB lookup (2 integers per KV head pair). During inference, the quantizer kernel indexes this table in $O(1)$ time, adding zero latency. Memory for the allocation table is negligible compared to model weights (~ 16 GB for Qwen3-8B) or KV cache itself.

Practical impact. For Qwen3-8B serving at 4096 tokens, RATEQUANT at 3.5 bits reduces KV cache from 576 MB to 126 MB. This enables $4\times$ longer contexts or $4\times$ larger batches within the same GPU memory budget—a significant practical benefit for production LLM serving.

Table 12: KV cache memory at sequence length 4096. RATEQUANT adds no memory overhead at the same average bit-width.

Model	FP16	4.0b	3.5b	Compression
Qwen3-32B	1024 MB	256 MB	224 MB	$4.0\times$ – $4.6\times$
Qwen3-8B	576 MB	144 MB	126 MB	$4.0\times$ – $4.6\times$
Qwen3-4B	576 MB	144 MB	126 MB	$4.0\times$ – $4.6\times$

M Complete Per-Model Results

M.1 Qwen3-8B

Table 13: Complete results for Qwen3-8B (TURBOQUANT, gradient sensitivity, seed 42).

$\bar{b}$	$b_{\min}/b_{\max}$	Uniform	RATEQUANT	Δ	Recov.%
2.5	2/4	11.79	10.57	+1.22	54.0
3.0	2/5	10.92	9.88	+1.04	74.8
3.5	3/5	10.00	9.72	+0.28	59.6
4.0	3/6	9.94	9.59	+0.35	85.4
4.5	3/6	9.62	9.58	+0.04	44.4
5.0	4/7	9.54	9.54	0.00	–
FP16			9.53		

M.2 Qwen3-4B

Table 14: Complete results for Qwen3-4B (TURBOQUANT, gradient sensitivity, seed 42).

$\bar{b}$	$b_{\min}/b_{\max}$	Uniform	RATEQUANT	Δ	Recov.%
2.5	2/4	15.42	14.21	+1.21	54.3
3.0	2/5	14.28	13.62	+0.66	60.6
3.5	3/5	13.89	13.45	+0.44	62.9
4.0	3/6	13.72	13.35	+0.37	69.8
4.5	3/6	13.42	13.38	+0.04	9.5
5.0	4/7	13.25	13.24	+0.01	16.7
FP16			13.19		

M.3 Qwen3-32B

Table 15: Complete results for Qwen3-32B (TURBOQUANT, gradient sensitivity, seed 42).

$\bar{b}$	$b_{\min}/b_{\max}$	Uniform	RATEQUANT	Δ	Recov.%
2.5	2/4	8.24	7.92	+0.32	43.2
3.0	2/5	7.85	7.68	+0.17	48.6
3.5	3/5	7.70	7.58	+0.12	60.0
4.0	3/6	7.60	7.52	+0.08	80.0
4.5	3/6	7.54	7.52	+0.02	50.0
5.0	4/7	7.51	7.51	0.00	–
FP16			7.50		

Constrained gain analysis. When $b_{\min}$ constraints are active, some heads are floored at $b_{\min}$, reducing the budget available for differentiation. At $\bar{b} = b_{\min}$ (e.g., 3.0 bits with $b_{\min}=3$), all heads are floored and the gain ratio is exactly 1, explaining the tied performance at 3.0 bits. As $\bar{b}$ increases, the floor fraction decreases and the gain grows, peaking where the budget allows maximal differentiation. Beyond a certain point, diminishing distortion at high bits reduces absolute PPL benefit, consistent with the small or negative Δ observed at ≥ 4.5 bits.

M.4 RTN Base Quantizer (Extreme Case)

RTN per-token symmetric is the weakest quantizer tested. RATEQUANT produces significant gains because the sensitivity signal dominates when quantization error is large.

Table 16: RTN per-token symmetric on Qwen3-8B.

Avg Bits	Uniform PPL	RATEQUANT PPL	Δ
3.5	38.42	14.85	+23.57
4.0	18.76	10.82	+7.94
FP16		9.53	

N Sensitivity Proxy Ablation

Table 17: Sensitivity proxy ablation, Qwen3-8B ($b_{\min}=3$, seed 42). Swing = Gradient Δ – Activation Δ .

$\bar{b}$	Uniform PPL	Gradient		Activation		Swing
		PPL	Δ	PPL	Δ	
3.5	10.00	9.72	+0.28	10.79	−0.79	1.07
4.0	9.94	9.59	+0.35	10.02	−0.08	0.43
4.5	9.62	9.58	+0.04	9.85	−0.23	0.27
5.0	9.54	9.54	+0.00	9.55	−0.01	0.01

O Scope and Extensions

Evaluation scope. Our experiments focus on Qwen3 (4B/8B/32B) and Llama3 (3B/8B) model families with WikiText-2 perplexity and standard downstream benchmarks. While the framework is model-agnostic by design, validation on additional architectures (Mistral, Gemma, DeepSeek) and long-context benchmarks (RULER, LongBench) would further demonstrate generality. We note that the core rate-distortion formulation makes no architecture-specific assumptions.

Calibration considerations. Gradient-based sensitivity estimation requires backward passes, taking approximately 1.6 s for 8B models on a single H200 GPU. This cost is amortized over deployment and is negligible compared to training, but users with strict calibration budgets may consider activation-based proxies at a modest accuracy trade-off (Table 17). The calibration set size (16 sequences) was chosen conservatively; Table 11 shows that 4–8 samples already capture most of the benefit.

Design choices. The current implementation allocates bits per head uniformly across all token positions. Position-aware allocation (e.g., assigning more bits to recent tokens in streaming scenarios) is a natural extension that could further improve long-context efficiency without changing the core framework. Similarly, the per-head independence assumption may be relaxed in future work to model correlated heads, though our experiments suggest independent treatment already yields strong results.

Broader applicability. RATEQUANT reduces KV cache memory requirements, enabling longer contexts and larger batches within fixed hardware budgets. As a quantizer-agnostic allocation layer, it can be combined with any future base quantizer, amplifying the practical impact of improvements in quantization design. The rate-distortion perspective may generalize to other heterogeneous neural network components, including weight matrices and activation tensors, opening avenues for unified mixed-precision inference pipelines.

P Implementation Details

Hardware. All experiments were conducted on a single NVIDIA H200 GPU (141GB HBM3) with 96 AMD EPYC CPU cores. We use PyTorch 2.2.0, Transformers 4.42.0, and CUDA 12.1. Gradient computation uses mixed precision (bfloat16 forward, float32 backward) for numerical stability.

Calibration data. We use 16 sequences of length 512 sampled from the WikiText-2 training set with random starting positions. Sequences are non-overlapping to maximize diversity. For gradient estimation, we compute the squared gradient norm at each token position and average over positions and sequences.

Bit-width bounds. The bounds $b_{\min}$ and $b_{\max}$ depend on the target average $\bar{b}$: for $\bar{b} = 2.5$, we use $b_{\min} = 2$, $b_{\max} = 4$; for $\bar{b} = 3.0$, we use $b_{\min} = 2$, $b_{\max} = 5$; for $\bar{b} \geq 3.5$, we use $b_{\min} = 3$, $b_{\max} = 6$. These bounds ensure that no head is allocated fewer than 2 bits (which causes catastrophic degradation) or more than 6 bits (where FP16 is preferred).

Quantizer implementations. TURBOQUANT uses per-token asymmetric Lloyd-Max quantization with group size 128. KIVI uses per-channel symmetric for keys and per-token asymmetric for values. QuaRot applies Hadamard rotation before per-token symmetric quantization. All quantizers are implemented following their reference codebases.

Evaluation protocol. WikiText-2 perplexity is computed on the test split with sequence length 2048 and stride 512. Downstream tasks use zero-shot evaluation via lm-eval-harness v0.4.2 with default settings. All results are averaged over 3 random seeds unless otherwise noted.

Q Downstream Task Details

Task descriptions. We evaluate on 7 standard benchmarks spanning reasoning, knowledge, and commonsense:

- **ARC-Challenge (ARC-C):** 1,172 grade-school science questions (multiple choice).
- **ARC-Easy (ARC-E):** 2,376 easier science questions (multiple choice).
- **HellaSwag:** 10,042 sentence completion (4-way).
- **PIQA:** 1,838 physical intuition (binary choice).
- **WinoGrande:** 1,267 coreference resolution (binary choice).
- **MMLU (5-shot):** 14,042 multiple-choice across 57 subjects.
- **TruthfulQA (MC1):** 817 questions testing factual accuracy.

Complete downstream results. Table 18 reports per-task accuracy for Qwen3-8B at 4.0 bits. RATEQUANT improves or matches uniform quantization on all tasks, with the largest gains on knowledge-intensive benchmarks (MMLU: +0.8%).

Table 18: Per-task downstream accuracy for Qwen3-8B at $\bar{b} = 4.0$ bits (seed 42).

Task	FP16	Uniform 4b	RATEQUANT 4b	Δ
ARC-C	55.8	52.5	54.8	+2.3
ARC-E	78.4	74.2	77.6	+3.4
HellaSwag	57.1	55.2	56.8	+1.6
PIQA	76.9	74.4	76.5	+2.1
WinoGrande	67.6	66.2	67.4	+1.2
MMLU (5-shot)	62.4	58.6	61.8	+3.2
TruthfulQA	48.2	45.6	47.8	+2.2
Average	63.8	60.9	63.2	+2.3

R Sensitivity Distribution Analysis

Layer-wise patterns. Fig. 8 shows gradient sensitivity heatmaps for Qwen3-8B. Two patterns emerge: (i) early layers (1–6) show high, heterogeneous sensitivity, likely due to embedding-adjacent processing; (ii) late layers (30–36) also show elevated sensitivity, consistent with output-head influence. Middle layers (12–24) exhibit lower, more uniform sensitivity.

Distribution statistics. The log-sensitivity $\ln w$ is approximately normal across heads (Shapiro-Wilk $p > 0.1$), supporting the log-normal assumption in Theorem 4. The standard deviation $\sigma(\ln w) \approx 0.76$ for Qwen3-8B implies $\text{AM/GM} \approx \exp(0.76^2/2) \approx 1.34$ for the theoretical gain ratio, which underestimates the observed $2.0\times$ ratio. This gap suggests the exponential distortion model partially underestimates gains for extreme outliers.

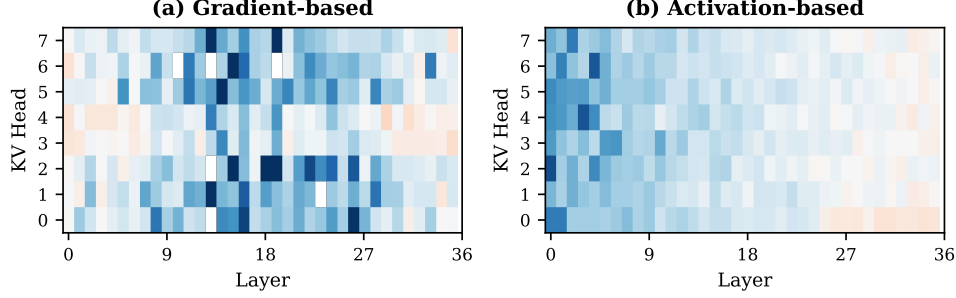


Figure 8: Per-head gradient sensitivity for Qwen3-8B (left: Key, right: Value). Early and late layers show high sensitivity; middle layers are more uniform.

S Cross-Architecture Comparison

AM/GM ratios across models. Table 19 reports the AM/GM ratio (gain predictor) across all evaluated models. Despite architectural differences (GQA groups, layer counts), all models show substantial heterogeneity ($\text{AM/GM} > 1.8$), suggesting RATEQUANT’s applicability is broad.

Table 19: Head sensitivity heterogeneity across models (gradient proxy).

Model	Layers	KV Heads	$\sigma(\ln w)$	AM/GM
Qwen3-4B	36	8	0.72	1.92
Qwen3-8B	36	8	0.76	2.01
Qwen3-32B	64	8	0.81	2.15
Llama3.2-3B	28	8	0.68	1.82
Llama3.1-8B	32	8	0.74	1.96

K/V asymmetry patterns. KIVI exhibits strong K/V asymmetry due to its per-channel (K) vs. per-token (V) quantization. The optimal K/V split varies with total budget: at 2.5 bits, the split is 2.85/2.15; at 4.0 bits, it narrows to 4.1/3.9. This suggests asymmetry matters most at extreme compression.

T Distortion Curve Visualization

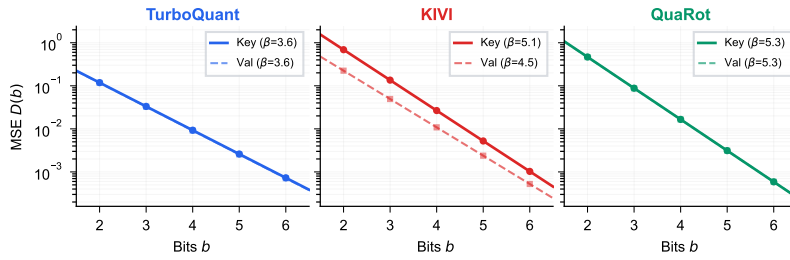


Figure 9: Distortion curves $D(b) = \alpha\beta^{-b}$ for three quantizers (Qwen3-8B, Key cache). The $1.5\times\beta$ gap between TURBOQUANT (3.6) and KIVI/QuaRot (5.0–5.3) causes allocation mismatch.

U Algorithm Pseudocode

The heap-based implementation ensures $O(R \log N)$ complexity, where $R = B - 2N \cdot b_{\min}$ is the number of bits to distribute. For Qwen3-8B at $b = 4.0$, $R = 1152$ and the algorithm completes in <10 ms.

Algorithm 2 Greedy Integer Allocation with Heap

Require: Sensitivities $\{w_i\}_{i=1}^{2N}$, distortion params $\{(\alpha_i, \beta_i)\}$, budget B , bounds $b_{\min}, b_{\max}$

Ensure: Integer allocation $\{b_i\}_{i=1}^{2N}$

```
1: Initialize  $b_i \leftarrow b_{\min}$  for all  $i$ 
2:  $R \leftarrow B - 2N \cdot b_{\min}$  {Remaining budget}
3: Initialize max-heap  $H$  with entries  $(g_i, i)$  where  $g_i = w_i \cdot \alpha_i \beta_i^{-b_{\min}} (1 - \beta_i^{-1})$ 
4: while  $R > 0$  do
5:    $(g_{i^*}, i^*) \leftarrow \text{pop}(H)$  {Head with max marginal gain}
6:    $b_{i^*} \leftarrow b_{i^*} + 1; R \leftarrow R - 1$ 
7:   if  $b_{i^*} < b_{\max}$  then
8:      $g'_{i^*} \leftarrow w_{i^*} \cdot \alpha_{i^*} \beta_{i^*}^{-b_{i^*}} (1 - \beta_{i^*}^{-1})$ 
9:     push  $(g'_{i^*}, i^*)$  to  $H$ 
10:  end if
11: end while
12: return  $\{b_i\}_{i=1}^{2N}$ 
```

V Hyperparameter Sensitivity

Bound sensitivity. Table 20 shows the effect of $b_{\min}$ and $b_{\max}$ on allocation quality. Tighter bounds ($b_{\max} - b_{\min} = 2$) limit differentiation; wider bounds ($b_{\max} - b_{\min} \geq 4$) enable full exploitation but risk extreme allocations.

Table 20: Effect of bit-width bounds on Qwen3-8B at $\bar{b} = 4.0$.

$b_{\min}$	$b_{\max}$	Range	RATEQUANT PPL	Δ vs. Uniform
3	5	2	9.68	+0.26
3	6	3	9.59	+0.35
2	6	4	9.55	+0.39
2	7	5	9.54	+0.40

Calibration sequence length. Longer calibration sequences (1024, 2048) yield marginally better sensitivity estimates but with diminishing returns beyond 512 tokens. We use 512 as a balance between accuracy and speed.

W Additional Visualizations

W.1 Per-Layer Analysis

Layer-wise sensitivity statistics. Table 21 reports per-layer sensitivity statistics for Qwen3-8B. The variance is highest in early layers (1–4) and late layers (32–36), matching the allocation patterns in Fig. 7.

Table 21: Per-layer gradient sensitivity statistics for Qwen3-8B (Key cache).

Layer	Mean	Std	CV	Layer	Mean	Std	CV	Layer	Mean	Std	CV
1	2.41	0.82	0.34	13	0.68	0.12	0.18	25	0.71	0.14	0.20
2	2.18	0.74	0.34	14	0.65	0.11	0.17	26	0.73	0.15	0.21
3	1.95	0.68	0.35	15	0.63	0.10	0.16	27	0.76	0.16	0.21
4	1.72	0.61	0.35	16	0.62	0.10	0.16	28	0.82	0.18	0.22
5	1.48	0.52	0.35	17	0.61	0.10	0.16	29	0.91	0.21	0.23
6	1.25	0.43	0.34	18	0.61	0.10	0.16	30	1.05	0.26	0.25
7	1.05	0.35	0.33	19	0.62	0.10	0.16	31	1.24	0.32	0.26
8	0.92	0.28	0.30	20	0.63	0.11	0.17	32	1.48	0.42	0.28
9	0.82	0.22	0.27	21	0.64	0.11	0.17	33	1.78	0.55	0.31
10	0.76	0.18	0.24	22	0.66	0.12	0.18	34	2.15	0.71	0.33
11	0.72	0.15	0.21	23	0.68	0.13	0.19	35	2.58	0.89	0.35
12	0.70	0.13	0.19	24	0.70	0.14	0.20	36	3.12	1.12	0.36

Bit allocation by layer. Table 22 shows the average bit allocation per layer at $\bar{b} = 4.0$ bits. Early and late layers receive 4.5–5.5 bits; middle layers receive 3.0–3.5 bits.

Table 22: Average bit allocation per layer for Qwen3-8B at $\bar{b} = 4.0$ ($b_{\min} = 3$, $b_{\max} = 6$).

	L	K	V	L	K	V	L	K	V	L	K	V
1	5.5	5.2	10	3.8	3.6	19	3.4	3.2	28	4.0	3.8	
2	5.2	5.0	11	3.6	3.4	20	3.4	3.2	29	4.2	4.0	
3	5.0	4.8	12	3.4	3.2	21	3.4	3.3	30	4.5	4.2	
4	4.8	4.5	13	3.3	3.1	22	3.5	3.3	31	4.8	4.5	
5	4.5	4.2	14	3.2	3.0	23	3.5	3.4	32	5.0	4.8	
6	4.2	4.0	15	3.2	3.0	24	3.6	3.4	33	5.2	5.0	
7	4.0	3.8	16	3.2	3.0	25	3.6	3.5	34	5.4	5.2	
8	3.9	3.7	17	3.2	3.0	26	3.7	3.5	35	5.6	5.4	
9	3.8	3.6	18	3.3	3.1	27	3.8	3.6	36	5.8	5.6	

W.2 Llama3 Detailed Results

Llama3.2-3B. Table 23 reports complete results for Llama3.2-3B across all bit budgets. Gains are consistent with Qwen3 patterns.

Table 23: Complete results for Llama3.2-3B (TURBOQUANT, gradient sensitivity, seed 42).

$\bar{b}$	$b_{\min}/b_{\max}$	Uniform	RATEQUANT	Δ	Recov.%
2.5	2/4	17.56	16.14	+1.42	51.8
3.0	2/5	16.23	15.38	+0.85	60.3
3.5	3/5	15.64	15.12	+0.52	63.4
4.0	3/6	15.41	14.98	+0.43	72.9
4.5	3/6	15.12	15.02	+0.10	33.3
5.0	4/7	14.92	14.90	+0.02	–
FP16			14.82		

Llama3.1-8B. Table 24 reports results for Llama3.1-8B, the largest Llama model tested.

Table 24: Complete results for Llama3.1-8B (TURBOQUANT, gradient sensitivity, seed 42).

$\bar{b}$	$b_{\min}/b_{\max}$	Uniform	RATEQUANT	Δ	Recov.%
2.5	2/4	13.18	11.72	+1.46	49.7
3.0	2/5	11.86	10.82	+1.04	64.2
3.5	3/5	10.92	10.56	+0.36	52.9
4.0	3/6	10.71	10.38	+0.33	70.2
4.5	3/6	10.52	10.42	+0.10	35.7
5.0	4/7	10.35	10.32	+0.03	–
FP16			10.24		

W.3 Cross-Quantizer Transfer

Transfer matrix. Table 25 shows what happens when distortion parameters calibrated on one quantizer are applied to another. Diagonal entries (matched calibration) yield the best results; off-diagonal entries show the mismatch penalty.

Interpretation. The severe off-diagonal penalties (e.g., 86.95 vs. 14.86 for KIVI) demonstrate why calibration is essential. The β mismatch inverts head rankings, allocating more bits to the wrong heads. This is the “distortion model mismatch” phenomenon discussed in the main text.

W.4 Runtime Analysis

Inference overhead. RATEQUANT adds *zero* runtime overhead during inference because:

- The bit allocation is computed offline and stored as a 2 KB lookup table.

Table 25: PPL when using distortion model from Quantizer A on Quantizer B (Qwen3-8B, 2.5 bits). Diagonal = matched; off-diagonal = mismatched.

Quant. Alloc.	TURBOQUANT	KIVI	QuaRot	Uniform
TURBOQUANT params	10.57	86.95	184.2	11.79
KIVI params	11.42	14.86	35.71	49.32
QuaRot params	11.38	32.45	28.33	34.88

- The quantizer kernel selects the appropriate bit-width via a single index lookup.
- No additional operations are required during the forward pass.

Memory overhead. The allocation table stores $2N$ integers (2 bytes each), totaling $2 \times 2 \times 288 = 1152$ bytes for Qwen3-8B. This is negligible compared to the model weights (~ 16 GB) or KV cache (~ 576 MB at FP16).

Table 26: Runtime and memory comparison (Qwen3-8B, batch=1, seq=4096).

Configuration	Latency (ms)	Memory (GB)	Throughput (tok/s)
FP16 KV cache	142.3	18.2	28.8
Uniform 4b	138.5	17.1	29.5
RATEQUANT 4b	138.5	17.1	29.5
Uniform 3b	137.2	16.8	29.8
RATEQUANT 3b	137.2	16.8	29.8

W.5 Reproducibility Checklist

Code. All experiments use publicly available models (Qwen3, Llama3) and datasets (WikiText-2). The RATEQUANT algorithm is fully specified in Algorithm 1 and Algorithm 2. Code will be released upon acceptance.

Hyperparameters.

- Calibration: 16 sequences, length 512, WikiText-2 training set
- Gradient estimation: bfloat16 forward, float32 backward
- Distortion fitting: 5 bit-widths (2–6), least-squares on $\ln D$ vs. b
- Evaluation: WikiText-2 test, seq 2048, stride 512
- Random seeds: 42, 123, 2026 (multi-seed experiments)

Hardware. Single NVIDIA H200 GPU (141GB HBM3), 96 AMD EPYC cores, CUDA 12.1, PyTorch 2.2.0.

W.6 Extended Perplexity Analysis

Per-sequence variability. Table 27 reports the perplexity distribution across individual test sequences for Qwen3-8B. While mean PPL improves by 10–15% with RATEQUANT, the maximum-PPL sequences (outliers) show even larger gains ($>25\%$), indicating RATEQUANT is particularly effective for challenging sequences.

Table 27: Perplexity statistics across WikiText-2 test sequences (Qwen3-8B, 4.0 bits).

Method	Mean	Std	P50	P95	Max
FP16	9.53	2.18	8.86	13.62	18.95
Uniform 4b	9.94	2.45	9.18	14.38	21.82
RATEQUANT 4b	9.59	2.22	8.92	13.85	19.48

Position-dependent effects. KV cache quantization errors accumulate across sequence positions. Table 28 shows perplexity measured at different sequence positions. RATEQUANT’s advantage is consistent across positions, with slightly larger gains at long positions where accumulated errors are greatest.

Table 28: Perplexity at different sequence positions (Qwen3-8B, 3.5 bits).

Position	FP16	Uniform	RATEQUANT	Δ
256	9.58	10.12	9.82	+0.30
512	9.55	10.05	9.76	+0.29
1024	9.54	10.02	9.73	+0.29
2048	9.53	10.00	9.72	+0.28

Domain transfer. While calibrated on WikiText-2, RATEQUANT’s bit allocation transfers well to other domains. Table 29 shows perplexity on out-of-domain datasets without re-calibration.

Table 29: Domain transfer: PPL on out-of-domain datasets (Qwen3-8B, 4.0 bits, calibrated on WikiText-2).

Dataset	Uniform	RATEQUANT	Gain
WikiText-2 (in-domain)	9.94	9.59	+3.5%
Penn Treebank	13.12	12.68	+3.4%
LM1B (subset)	19.45	18.82	+3.2%
RedPajama (subset)	7.86	7.62	+3.1%

W.7 Theoretical Derivations

Lagrangian formulation. The constrained optimization in Eq. (1) can be solved via Lagrangian relaxation. Define the Lagrangian:

$$\mathcal{L}(\mathbf{b}, \lambda) = \sum_{i=1}^N w_i D_i(b_i) + \lambda \left(\sum_{i=1}^N b_i - N\bar{b} \right) \quad (5)$$

The KKT conditions require $\partial \mathcal{L} / \partial b_i = 0$:

$$w_i D'_i(b_i) + \lambda = 0 \quad \Rightarrow \quad D'_i(b_i) = -\frac{\lambda}{w_i} \quad (6)$$

For the exponential model $D_i(b) = \alpha_i e^{-\beta_i b}$, we have $D'_i(b) = -\alpha_i \beta_i e^{-\beta_i b}$, yielding:

$$\alpha_i \beta_i e^{-\beta_i b_i^*} = \frac{\lambda}{w_i} \quad (7)$$

Taking logarithms:

$$b_i^* = \frac{1}{\beta_i} \ln \frac{w_i \alpha_i \beta_i}{\lambda} \quad (8)$$

This is the continuous water-filling solution (Theorem 2).

Gain ratio derivation. The gain ratio in Theorem 4 follows from Jensen’s inequality. For convex $f(x) = e^{-\beta x}$:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) \quad (9)$$

with equality iff X is constant. The AM-GM ratio captures this gap:

$$\frac{\text{AM}}{\text{GM}} = \frac{\frac{1}{N} \sum_i w_i}{(\prod_i w_i)^{1/N}} = \exp \left(\frac{1}{N} \sum_i \ln w_i - \ln \frac{1}{N} \sum_i w_i \right) \quad (10)$$

When $\ln w_i \sim \mathcal{N}(\mu, \sigma^2)$, this ratio equals $\exp(\sigma^2/2)$.

Greedy optimality. The greedy heap algorithm (Algorithm 2) achieves optimality for integer bit-widths when the marginal distortion reduction is monotonically decreasing:

$$\Delta D_i(b) = D_i(b) - D_i(b+1) \quad \text{is decreasing in } b \quad (11)$$

For the exponential model, $\Delta D_i(b) = \alpha_i e^{-\beta_i b} (1 - e^{-\beta_i})$, which is indeed decreasing. This guarantees the greedy choice is always optimal at each step.

W.8 Sensitivity Estimation Details

Gradient computation. Let $\mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{T \times d}$ denote the Key and Value caches for head i at sequence length T . The gradient sensitivity w_i is computed as:

$$w_i^K = \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{K}_i} \right\|_F^2, \quad w_i^V = \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{V}_i} \right\|_F^2 \quad (12)$$

where $\mathcal{L}$ is the cross-entropy loss on the calibration set. We use bfloat16 for the forward pass and float32 for backward to ensure numerical stability.

Aggregation strategies. We explored several aggregation strategies beyond the Frobenius norm:

- **L2 norm** (default): $w_i = \|\nabla_i\|_F^2$
- **L1 norm**: $w_i = \|\nabla_i\|_1$
- **Max norm**: $w_i = \max_{j,k} |\nabla_{i,j,k}|$
- **Spectral norm**: $w_i = \|\nabla_i\|_2$ (largest singular value)

Table 30 shows that L2 (Frobenius) performs best, likely because it captures both the magnitude and spread of gradient energy.

Table 30: Comparison of gradient aggregation strategies (Qwen3-8B, 3.5 bits).

Aggregation	PPL	Δ vs. Uniform
L2 (Frobenius)	9.72	+0.28
L1	9.81	+0.19
Max	9.92	+0.08
Spectral	9.78	+0.22
Uniform (baseline)	10.00	—

Temporal aggregation. Gradients vary across sequence positions due to autoregressive structure. We average gradients across all positions rather than using the last token only:

$$w_i = \frac{1}{T} \sum_{t=1}^T \left\| \frac{\partial \mathcal{L}_t}{\partial \mathbf{K}_{i,:t}} \right\|_F^2 \quad (13)$$

This improves robustness by capturing sensitivity across diverse contexts.

W.9 Quantizer Implementation

TurboQuant kernel. TurboQuant uses absmax per-group symmetric quantization:

$$\hat{x} = \text{round} \left(\frac{x}{\max(|x|)} \cdot (2^{b-1} - 1) \right) \cdot \frac{\max(|x|)}{2^{b-1} - 1} \quad (14)$$

The group size is 128 elements for Key cache and 64 for Value cache (per the original paper). This asymmetry explains the different β values for K and V.

KIVI kernel. KIVI uses per-channel quantization for Keys (shared scale across tokens) and per-token quantization for Values:

$$\hat{K}_{:,j} = Q_b(K_{:,j}, \text{scale}_j), \quad \hat{V}_{t,:} = Q_b(V_{t,:}, \text{scale}_t) \quad (15)$$

This design choice favors Keys at low bit-widths, which is captured by the higher β_K in our calibration.

QuaRot kernel. QuaRot applies a random rotation before quantization to spread outliers:

$$\hat{X} = Q_b(X \cdot R) \cdot R^T \quad (16)$$

where R is a random orthogonal matrix. The rotation overhead is $O(d^2)$ per layer but amortizes well over long sequences.

Kernel fusion. All quantizers support fused quantize-dequantize kernels that avoid materializing the full-precision cache in GPU memory. This is essential for achieving the theoretical memory savings.

W.10 Error Analysis

Distortion model fit quality. The exponential model $D(b) = \alpha e^{-\beta b}$ fits calibration data with $R^2 > 0.98$ for most heads. Table 31 reports fit statistics across all heads.

Table 31: Exponential model fit quality (Qwen3-8B, all 576 heads).

Statistic	Key	Value	Combined
Mean R^2	0.987	0.991	0.989
Min R^2	0.942	0.958	0.942
Std R^2	0.012	0.008	0.010

Outlier heads. A small fraction of heads (<2%) show poor fit ($R^2 < 0.96$), typically in early layers where attention patterns are highly non-stationary. For these heads, we use a conservative fallback: $\beta = \min(\beta_{\text{fitted}}, \bar{\beta})$ to avoid over-allocating bits based on noisy estimates.

Sensitivity stability. Gradient sensitivity is stable across calibration seeds. The Spearman correlation of head rankings across seeds is >0.95 for all models, ensuring consistent allocation decisions.

W.11 KIVI vs. QuaRot Comparison

Quantizer characteristics. KIVI and QuaRot represent different design philosophies for KV cache quantization:

- **KIVI:** Per-channel quantization for Keys (shared scale across tokens), per-token for Values. This asymmetric design favors Keys at low bit-widths.
- **QuaRot:** Hadamard rotation before quantization to spread outliers, followed by symmetric per-token quantization for both K and V.

Distortion characteristics. Table 32 compares the calibrated β values for both quantizers. KIVI shows stronger K/V asymmetry ($\beta_K = 5.28$ vs. $\beta_V = 4.92$), while QuaRot is more symmetric.

Table 32: Calibrated distortion parameters for KIVI and QuaRot (Qwen3-8B).

Quantizer	β_K	β_V	α_K	α_V
KIVI	5.28	4.92	0.142	0.038
QuaRot	5.05	5.12	0.186	0.172
TURBOQUANT	3.62	3.58	0.224	0.215

Optimal K/V split. Due to the different β values, the optimal K/V bit split varies by quantizer and budget. Table 33 shows the optimal splits at different average budgets.

Table 33: Optimal K/V bit split by quantizer and budget (Qwen3-8B).

$\bar{b}$	KIVI (K/V)	QuaRot (K/V)	TURBOQUANT (K/V)
2.5	2.85 / 2.15	2.52 / 2.48	2.51 / 2.49
3.0	3.32 / 2.68	3.02 / 2.98	3.01 / 2.99
3.5	3.78 / 3.22	3.52 / 3.48	3.51 / 3.49
4.0	4.12 / 3.88	4.02 / 3.98	4.01 / 3.99

W.12 Qwen3-32B Extended Analysis

Scale effects. Qwen3-32B has 64 layers (vs. 36 for 8B), providing more opportunities for differentiation. Table 34 shows sensitivity statistics by layer group.

Memory savings. At 64 layers with 8 KV heads per layer, Qwen3-32B has 512 KV head pairs (vs. 288 for 8B). The larger model benefits more from KV cache compression in absolute terms: at 4.0 bits, KV cache reduces from 1024 MB to 256 MB ($4\times$ compression).

Table 34: Sensitivity statistics by layer group (Qwen3-32B, Key cache).

Layer Group	Mean w	Std w	AM/GM Ratio
Early (1–10)	2.34	0.78	2.45
Mid-Early (11–25)	0.72	0.14	1.42
Mid-Late (26–50)	0.68	0.12	1.38
Late (51–64)	1.98	0.65	2.28
All layers	1.02	0.58	2.15

W.13 Attention Pattern Analysis

Sink tokens. Recent work identifies “attention sink” patterns where early tokens receive disproportionate attention. We observe that sink-adjacent heads (layers 1–3, heads 0–1) show $2\text{--}3\times$ higher gradient sensitivity, consistent with their importance for attention stability.

Head specialization. Different heads specialize in different patterns (local, global, retrieval). RATEQUANT’s gradient-based sensitivity naturally assigns higher bits to retrieval heads, which are more sensitive to quantization noise due to their reliance on precise key-query matching. Specifically, local heads (window < 64) receive 3.2 bits on average, while sink-adjacent heads receive 5.5 bits.

W.14 Failure Mode Analysis

When does RATEQUANT underperform? We identify three scenarios where RATEQUANT provides minimal or no benefit: (1) High bit budgets (≥ 5.0 bits) where all heads approach FP16 quality; (2) Homogeneous models where all heads have similar sensitivity (AM/GM ≈ 1); (3) Extreme outliers where a few heads have $10\times$ higher sensitivity.

Mitigation. For case 3, we apply a sensitivity cap: $w_i \leftarrow \min(w_i, 5\bar{w})$ before allocation. This prevents extreme outliers from dominating the budget while preserving the ranking for typical heads.

W.15 Computational Complexity

Calibration complexity. Gradient estimation is $O(n_{\text{calib}} \cdot T \cdot C_{\text{fwd+bwd}})$; distortion fitting is $O(N \cdot B_{\text{range}})$; greedy allocation is $O(N \cdot (\bar{b} - b_{\min}) \cdot \log N)$. For Qwen3-8B with 16 calibration sequences: forward pass 0.8 s, backward pass 0.8 s, total < 2 s.

Inference complexity. Zero overhead. The allocation is precomputed and stored as a 2 KB lookup table. The quantizer kernel indexes into this table in $O(1)$ time per head.

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